CalibAgent: Task-Aware Active Calibration and Shift Recovery for Quadruped Velocity Commands

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***Abstract*— Commercial and learned quadruped controllers often expose a planar velocity command while hiding the dynamics that map commands to realized motion. Payload, surface, controller, and estimator changes can therefore preserve balance yet distort navigation. We present CALIBAGENT, an active calibration layer for this opaque interface. A Bayesian command-to-motion model represents cross-axis coupling and asymmetric response, and a task-weighted integrated variance objective selects commands for a declared deployment distribution. Hard authorization, validation stopping, bounded inverse compensation, and residual-triggered recovery remain separate from acquisition. On 183 passive Unitree Go2 trials, a coupled affine map reduces held-out root-mean-square error (RMSE) by 54.5% relative to raw commands. Across controlled synthetic families, task weighting reaches a joint accuracy–uncertainty target in 18.67 trials, saving 39.5% over Latin hypercube sampling and 18.8% over D-optimal design. A candidate-noise oracle sensitivity changes the heteroscedastic result by one trial without changing the method ranking. Pinned Isaac Lab experiments show 9.3–31.7% held-out RMSE reduction and a matched-budget task-error advantage over D-optimal design on every navigation map. A paired residual-signature monitor produces 0/120 stationary-sequence alarms over 12,000 trials; across four shifts and 144 seeds per shift, detection is 99.3–100%, recovery is 97.9–100%, and active selection improves early recovery over passive updating. Failure-aware navigation is noninferior to a dense reference across six held-out maps. These results establish the active mechanism in controlled simulation and the need for a coupled response model in passive hardware data.**

I. INTRODUCTION

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\boldsymbol{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. Taouil *et al.* showed that learned motion models of a black-box quadruped controller can improve footstep planning [1]; related work has calibrated velocity-controlled kinematic models online for wheeled visual–inertial odometry [2]. The remaining question is how to collect the calibration trials themselves when hardware time is limited and the useful part of command space is task dependent.

Robot calibration has long treated measurement configuration as a design variable [3], [4]. Classical observability and D-optimal criteria prioritize parameter identification [5], [6]. Task-oriented calibration instead scores a design by uncertainty at the end-effector or another quantity of interest [7], [8]. We apply that principle to the external velocity interface of a fixed quadruped controller: trials are chosen for their reduction of predictive uncertainty over commands expected during deployment, not for uniform parameter-space coverage.

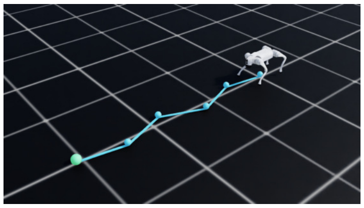
This layer differs from locomotion-policy adaptation. Rapid Motor Adaptation conditions a trained policy on latent environment estimates [9], residual models can update model-based legged controllers online [10], and recent embodiment adaptation identifies hardware changes from short interaction histories [11]. CALIBAGENT neither retrains nor instruments the low-level controller. It identifies the mapping from user commands to measured steady motion, translates desired velocities through a constrained inverse, and reactivates calibration when predictive residuals indicate a shift.

This paper makes four contributions. First, it formulates the opaque command-to-motion interface as a compact Bayesian regression with coupled, asymmetric features and per-trial measurement uncertainty. Second, it gives the conditional one-step task-weighted integrated variance reduction (IVR) score and a safe finite-pool sequential design. Third, it composes acquisition with hard authorization, validation stopping, constrained inversion, and latched shift recovery without allowing statistical value to relax an execution limit. Fourth, it evaluates the complete evidence chain from physical model need to downstream navigation using passive Go2 data, controlled synthetic families, fault injection, pinned physics simulation, and held-out navigation across six disjoint maps. The methodological contribution is this task-measure-to-command design and its closed-loop operational composition; the regression, optimal-design update, and change detector remain individually recognizable components. Hardware evidence in this study measures model necessity from passive trials; active selection, recovery, and navigation are evaluated in controlled simulation.

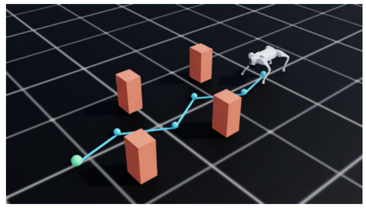
II. RELATED WORK

A. Calibration and Experimental Design

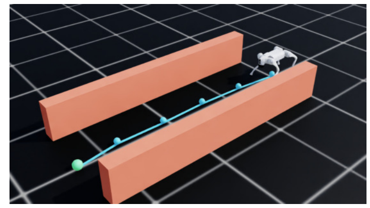
Kinematic and dynamic robot calibration depend on both model structure and the poses or trajectories used for identification [3], [4]. Sun and Hollerbach linked robot observability indices to alphabetic optimal design and developed an efficient active pose-selection algorithm [5], [6]. Carrillo *et al.* then selected configurations by predicted task-space error rather than only parameter covariance [7]. This task-space view parallels goal-oriented Bayesian design, which minimizes uncertainty in a downstream quantity of interest [8]. Greedy information selection also has established



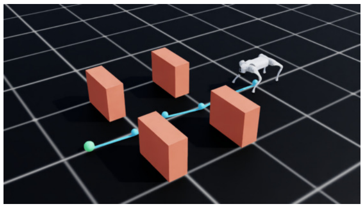
(a) S-bend



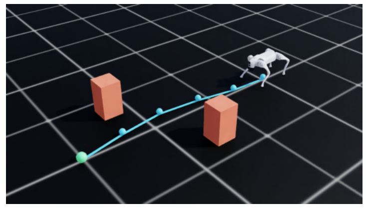
(b) Offset slalom



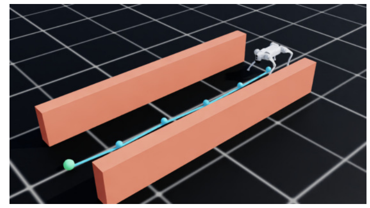
(c) Narrow lane



(d) Double chicane



(e) Weighted arc



(f) Extended lane

Fig. 1. Elevated Isaac Sim views of the six held-out navigation maps. Cyan segments and spheres show the map waypoints; orange geometry shows the obstacles. The overlays are non-colliding visualization geometry and do not contribute to the quantitative endpoints.

connections to near-optimal sensor placement [12]. CALIBAGENT uses a related predictive objective, but its design variable is a safety-filtered velocity command, its integration measure is the planner-facing command distribution, and its posterior is consumed by the same bounded inverse and change detector. In optimal-design terminology, Eq. (4) is a sequential Bayesian I/V-optimal criterion specialized to a multi-output command interface. Its contribution is the task-measure-to- command specialization, the separation of statistical selection from hard authorization, and the end-to-end calibration–recovery evidence.

B. Legged Command Models and Adaptation

Simulation-trained policies have produced agile and terrain-robust quadruped locomotion [13], [14], [15]. Robust locomotion, however, does not make the user-level velocity interface exact. Learned black-box motion models can bridge that interface for planning [1], while steady-state and local dynamics models can adapt a legged robot to unmodeled disturbances [16]. Policy-level adaptation and domain randomization address broader changes [9], [17], [11]. CALIBAGENT occupies a narrower operational layer: it actively calibrates a fixed controller's command response, quantifies posterior uncertainty, and applies compensation only inside a non-learned envelope.

III. METHOD

Figure 2 separates the probabilistic calibration loop from the rules that constrain its operation. The low-level locomotion controller is fixed and treated as a black box. A trial executes a constant body-frame command after warm-up and ramp-in, extracts a robust steady-motion observation, and records its covariance and validity. Invalid observations never update the posterior.

A. Bayesian Command-to-Motion Model

Let $\boldsymbol{y} \in \mathbb{R}^3$ denote the measured steady velocity for command $\boldsymbol{u}$. The coupled affine basis uses $[1, v_x, v_y, \omega_z]$; its diagonal counterpart retains the intercept but zeros every off-axis coefficient. The nonlinear coupled basis has 13 terms: the affine terms, three pairwise products, and $[v_k - \tau_k]_+, [-v_k - \tau_k]_+$ for each axis, with $\boldsymbol{\tau} = [0.15, 0.10, 0.25]$. The non-intercept columns are standardized once on a pre-declared safe command pool, never on observations or held-out outcomes. The affine basis is used for passive hardware comparison and navigation; the nonlinear basis is used for synthetic acquisition, closed-loop response, and shift recovery. Closed-loop fits initialize the linear projection to the identity interface and regularize deviations with a fixed prior. Consequently, the 12-trial navigation fit is a four-parameter-per-output affine update.

For output axis a, the model is

$$y_a = \boldsymbol{\theta}_a^\top \boldsymbol{\phi}(\boldsymbol{u}) + \epsilon_a, \quad \epsilon_a \sim \mathcal{N}(0, \sigma_a^2 + r_a), \tag{1}$$

where r_a is the measurement-pipeline covariance diagonal and σ_a^2 is the frozen process variance. Given prior $\mathcal{N}(\boldsymbol{m}_{a0}, \boldsymbol{\Sigma}_{a0})$, sequential updates use

$$\boldsymbol{\Lambda}_a \leftarrow \boldsymbol{\Lambda}_a + \frac{\boldsymbol{\phi}\boldsymbol{\phi}^\top}{\sigma_a^2 + r_a}, \tag{2}$$

$$\boldsymbol{\eta}_a \leftarrow \boldsymbol{\eta}_a + \frac{\boldsymbol{\phi} y_a}{\sigma_a^2 + r_a}, \quad \boldsymbol{\Sigma}_a = \boldsymbol{\Lambda}_a^{-1}, \quad \boldsymbol{m}_a = \boldsymbol{\Sigma}_a \boldsymbol{\eta}_a. \tag{3}$$

The predictive mean is $\mu_a(\boldsymbol{u}) = \boldsymbol{m}_a^\top \boldsymbol{\phi}(\boldsymbol{u})$, and the variance for a future noisy observation is $\boldsymbol{\phi}^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi} + \sigma_a^2 + r_a$; reported coverage uses this complete quantity, whereas acquisition uses only the reducible epistemic term. Independent output posteriors keep the online update small while coupled input features represent cross-axis response. Aggregate error uses $\|\boldsymbol{e}\|_W^2 = \boldsymbol{e}^\top W^\top W \boldsymbol{e}$ with $W = \mathrm{diag}((1\,\mathrm{m\,s^{-1}})^{-1}, (1\,\mathrm{m\,s^{-1}})^{-1}, (1\,\mathrm{rad\,s^{-1}})^{-1})$; this is numerically the declared equal-axis convention. Axiswise errors are reported alongside the scalar score.

B. Task-Weighted Active Selection

A declared task distribution is represented by grid commands $\{\boldsymbol{g}_j\}_{j=1}^G$ and normalized weights w_j. This grid lies in

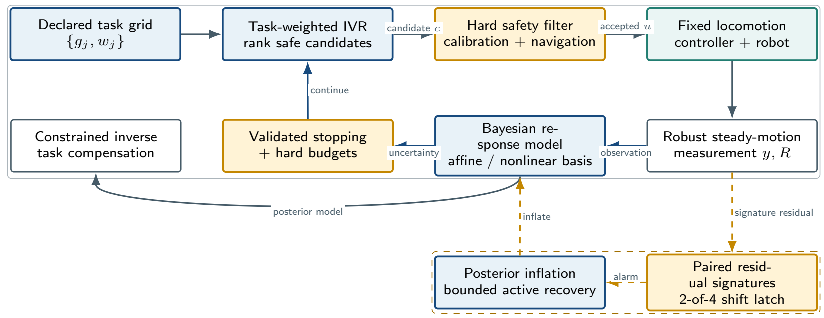


Fig. 2. CALIBAGENT architecture. Task-weighted acquisition ranks calibration commands, but the hard filter alone authorizes execution. Valid observations update the Bayesian model; held-out validation and uncertainty control stopping. A latched residual alarm inflates posterior uncertainty and opens a bounded active-recovery phase. The inverse compensator uses the same bounded command set during navigation.

the input domain of the forward response model. For navigation, nominal waypoint- planner outputs before compensation are a fixed proxy for the command inputs visited by the inverse solver; the support-mismatch experiment tests this proxy outside its declared range. For candidate $\boldsymbol{c}$, let $\tilde{r}_a(\boldsymbol{c})$ denote measurement variance available at planning time. A Sherman–Morrison update gives the outcome-independent, conditional one-step reduction in weighted epistemic variance:

$$\mathcal{I}(\boldsymbol{c}) = \sum_{a=1}^{3}\sum_{j=1}^{G} w_j \frac{\left(\boldsymbol{\phi}(\boldsymbol{g}_j)^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})\right)^2}{\sigma_a^2 + \tilde{r}_a(\boldsymbol{c}) + \boldsymbol{\phi}(\boldsymbol{c})^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})}. \quad (4)$$

The planner sets $\tilde{r}_a = 0$ because a future trial covariance is not observed before execution; Eq. (3) still uses its realized r_a. When a candidate-noise predictor is available, the implementation accepts its nonnegative $\tilde{r}_a(\boldsymbol{c})$. An oracle sensitivity supplies the known synthetic value. The planner maximizes $\mathcal{I}(\boldsymbol{c}) - \lambda_r R(\boldsymbol{c}) - \lambda_d D(\boldsymbol{c})$, where R is normalized command magnitude and D is normalized linear execution distance, over a bounded finite pool; the hard filter authorizes a candidate only after statistical ranking. The sample-efficiency experiment sets both coefficients to zero to isolate IVR. Greedy batches use covariance-only fantasy updates, so no unobserved target is imputed. The synthetic benchmark uses 256 candidates, a six-command seed design, a 256-point task grid, a disjoint 400-point evaluation grid, and a 0.025 normalized duplicate radius. Physics experiments instead use a hard-filtered 128- or 512-command pool and a 0.02 duplicate radius; their task measure is uniform over the held-out command set stated below.

C. Safety, Stopping, Compensation, and Shift Recovery

The hard filter checks finite state, localization, battery, roll, pitch, base height, command bounds, linear norm, coupled linear–angular load, command slew, and projected workspace occupancy. It evaluates ranked candidates after planning and fails closed when none remains. A 50 Hz monitor can zero the command independently of the planner. This separation prevents uncertainty reduction from trading against a learned safety penalty.

Stopping requires minimum trial coverage and measured held-out validation. A run stops at the declared target only after three consecutive checks of both validation RMSE and integrated uncertainty; a low-gain condition also requires valid held-out accuracy. Trial, time, distance, and battery budgets take priority. The synthetic benchmark separately computes an offline oracle crossing from simulator truth; that crossing is an evaluation endpoint and is never visible to acquisition. For deployment, the inverse compensator searches the same bounded candidate set and minimizes

$$\|W(\boldsymbol{\mu}(\boldsymbol{u}) - \boldsymbol{v}^*)\|_2^2 + \lambda\|W(\boldsymbol{u} - \boldsymbol{v}^*)\|_2^2 + \rho\sum_a s_a^2(\boldsymbol{u}), \quad (5)$$

then passes the selected command through the hard filter.

Shift monitoring cycles through four safe commands $(-.20, 0, -.20)$, $(-.20, 0, .20)$, $(.12, -.10, -.15)$, and $(.12, .10, .15)$. Commissioning freezes the corresponding residual signatures $\boldsymbol{r}_j^0 = \boldsymbol{y}_j - \boldsymbol{\mu}(\boldsymbol{u}_j)$. For a later response to signature command $j(t)$, the detector evaluates

$$d_t = \left\|\mathrm{diag}(.14, .08, .18)^{-1}\left(\boldsymbol{r}_t - \boldsymbol{r}_{j(t)}^0\right)\right\|_2. \quad (6)$$

An alarm requires $d_t > .30$ on at least two of the last four valid monitor trials and a two-trial minimum dwell, so an isolated response change cannot trigger recovery. The alarm latches, inflates epistemic covariance by a factor of eight without changing the posterior mean, and opens a 12-command recovery budget. Normal validation stopping resumes after recovery.

IV. EXPERIMENTAL EVALUATION

A. Passive Hardware and Sample Efficiency

The hardware dataset contains 183 valid passive Unitree Go2 trials collected in three 61-trial sessions. Livox MID360 FAST-LIO lidar–inertial odometry with global localization provides the reference pose. Each session is held out once; a training fold contains 122 trials, of which sampled fits use the first 30 commands of a fixed Latin hypercube design and dense fits use all 122. Each command uses 0.6 s ramp-in, 0.8 s settling, 2.0 s measurement, and 0.6 s ramp-out. Held-out three-axis and per-axis RMSE compare raw commands, diagonal affine calibration, and coupled affine calibration.

Additional diagonal-linear and coupled-linear fits separate intercept effects from cross-axis coupling.

Sample efficiency is evaluated over 20 independent seeds in each of three synthetic response families: affine, dead-zone, and heteroscedastic. Every nonlinear coupled fit begins with the same six-command seed design and has a 160-trial horizon. Candidate, task, and evaluation Sobol sets are independently scrambled. Commands span $[-1, 1] \times [-.5, .5] \times [-1.5, 1.5]$ with $\|(v_x, v_y)\|_2 \leq 1$. Ground truth is $\boldsymbol{y} = \text{clip}(A\boldsymbol{d}(\boldsymbol{u}) + \boldsymbol{b}, -\boldsymbol{s}, \boldsymbol{s}) + \boldsymbol{\epsilon}$: diagonal gains are uniform on $[.72, 1.08]$, off-axis terms on $[-.12, .12]$, and biases within $\pm(.04, .025, .06)$. The dead-zone and heteroscedastic families use $\boldsymbol{d} = \text{sign}(\boldsymbol{u})[|\boldsymbol{u}| - (.12, .08, .20)]_+$; the former clips at $(.82, .42, 1.20)$ and the other families at $(2, 1, 3)$. All families use base noise standard deviation $(.025, .018, .035)$; the heteroscedastic family multiplies it elementwise by $(1 + .7|\boldsymbol{u}|)$. The task measure is a Gaussian mixture centered at $(.55, 0, 0)$, $(.30, 0, .70)$, and $(.30, 0, -.70)$ with weights $(.50, .25, .25)$ and diagonal scales $(.14, .08, .18)$ for the first component and $(.12, .08, .16)$ for the turns. The model uses prior scale 1.0 and frozen process variances $(.000625, .000324, .001225)$. The 400-point evaluation grid is an offline ground-truth benchmark and never enters acquisition; an independent 1,024-command audit grid evaluates each posterior at its recorded crossing. The joint target is the first trial with RMSE at most 0.04 and integrated epistemic variance at most 0.0015. Controls comprise random, Latin hypercube (LHS), Sobol, Bayesian D-optimal, no-task IVR, and a 256-command dense oracle. Inference first averages the three response families within each paired seed. The one-sided paired Wilcoxon alternative is fewer trials for task IVR ($n = 20$ seeds; SciPy default zero handling). The LHS contrast is primary; all five baseline comparisons also survive Holm correction ($p_{\text{adj}} \leq 4.77 \times 10^{-6}$). Intervals for trials saved use 5,000 percentile resamples of paired seeds; fixed-budget error intervals use 4,000.

Task-distribution sensitivity freezes every acquisition and re-evaluates the resulting posteriors on a new 1,024-command Sobol grid. Besides the declared mixture, it changes the component weights to forward-heavy $(.8, .1, .1)$, left-heavy $(.1, .8, .1)$, and right-heavy $(.1, .1, .8)$, then replaces the mixture by a uniform measure over the entire safe envelope. No command is reselected. This distinguishes robustness to navigation-family reweighting from a support-expanding deployment change.

B. Closed-Loop Simulation and Safety Evaluation

All physics experiments use Isaac Lab v2.3.2, Isaac Sim 5.1.0, and PhysX [18]. Closed-loop response is measured in four contexts with 20 paired seeds per context. Each nonlinear coupled fit uses an identity-projection prior scale of 0.01, 12 calibration commands, and eight held-out validation commands. Its warm-up, ramp-in, settling, measurement, and ramp-out phases last 1.0, 0.4, 0.8, 1.0, and 1.0 s, respectively. The common eight-command task/validation measure is uniform over $(-.30, 0, \pm.35)$, $(.20, \pm.18, \pm.30)$ with all four sign combinations, and $(.35, 0, \pm.50)$.

For hardware and simulation, body-frame pose increments feed a robust constant-twist estimate. A Huber location gives steady velocity; the sample covariance of segment twists divided by the sample count supplies measurement covariance, whose diagonal enters Eq. (1). Timestamp, sample-count, command-constancy, and steady-window checks reject invalid observations.

The stopping and authorization tests replay 60 active-acquisition traces, exercise 300 out-of-envelope proposals and 20 safe controls, and inject 160 sampled-state faults. The tested bounds are $|v_x| \leq 0.75$ m/s, $|v_y| \leq 0.45$ m/s, and $|\omega_z| \leq 1.2$ rad/s, with a 20 ms monitoring period. Shift recovery uses two independently frozen 72-seed blocks for each of four held-out in-place changes (144 paired seeds per change): gain and coupling; friction, load, and gain; a mixed physical and nonlinear change; and load, center-of-mass (COM), and gain. The blocks share the method, endpoint, and gate definitions but use disjoint seed sets, simulator randomization, and post-shift draws. Fixed-model, passive-update, and CalibAgent arms share 12 initial calibration commands, four signature commissioning commands, five shifted monitor commands, a 12-command recovery budget, and four-command rolling validation. The early-recovery endpoint averages trials 4–9. A separate stationary-monitor experiment evaluates 30 sequences in each of four contexts after commissioning; every sequence has 100 monitor trials, giving 12,000 trials and 31,200 s of aggregate command time. A complementary selector ablation uses 30 paired seeds in each shift and holds the detector, covariance inflation, update rule, safety filter, and budget fixed while replacing task IVR with no-task IVR, D-optimal, LHS, or random selection. If no ranked candidate is authorized, every adaptive arm executes the same zero-command safe stop. The five trial phases last 0.5, 0.3, 0.6, 0.7, and 0.5 s.

C. Held-Out Navigation

Navigation uses six disjoint maps (Fig. 1), seven calibration methods, and 72 paired seeds per map, totaling 3,024 episodes. All arms share the same controller, 50 Hz monitor and predictive interlock, 10 Hz waypoint follower, map geometry, and randomization. The affine model's prior scale is 0.10 with process variances $(.0025, .0025, .0050)$; CalibAgent and matched controls use 12 calibration commands, whereas the dense reference uses 30. Eight fixed commands provide calibration validation. Task IVR uses a 512-command safe pool and a uniform measure over $(.12, 0, 0)$, $(.16, 0, 0)$, $(.18, 0, 0)$, $(.14, \pm.08, \pm.15)$, and $(.10, \pm.10, \pm.22)$, where paired signs are matched. The inverse uses regularization .005, posterior-risk weight .01, and axis confidence weights $(1, .5, .5)$.

Routes extend 2.85–3.60 m; the narrow corridors are 1.10–1.14 m wide, the smallest gate opening is 0.79 m, and collision tests use a 0.20 m footprint radius. These dimensions and the common dynamics randomization are fixed for all calibration arms.

Every navigation arm uses the same fixed 50 Hz predictive base-height interlock, fail-closed command filter, and 10 Hz waypoint planner.

The navigation command envelope is $v_x \in [-0.40, 0.40]$ m/s, $v_y \in [-0.30, 0.30]$ m/s, and $\omega_z \in [-0.70, 0.70]$ rad/s, with linear norm at most 0.45 m/s. Cruise, lateral, and yaw limits are 0.18 m/s, 0.14 m/s, and 0.25 rad/s. Success means entering the 0.25 m goal ball without collision or serious safety event before 60 s. Arrival time is defined only on success; the primary failure-aware time estimand assigns 60 s to every failure. Navigation time excludes calibration, which is reported separately: acquisition takes 31.2 s for a 12-trial method and 78.0 s for dense calibration. Including the common eight-command validation gives 52.0 and 98.8 s, respectively. Marginal rates use two-sided exact Clopper–Pearson intervals. Paired continuous endpoints, mean ratios, and paired success/collision differences use 4,000 percentile resamples of common seed indices. The navigation margins were fixed at a 1.25 capped-time ratio and a -0.05 success difference: the former limits excess time to 25% of a dense-calibrated traversal, and the latter limits loss to approximately four of 72 episodes when the reference is perfect. Supplementary code and data provide the configurations, metric tables, evaluators, figure builders, and commands needed to reproduce every reported result.

V. Results

A. Model Need and Calibration Efficiency

Figure 5a shows a consistent held-out benefit from coupled calibration on the passive Go2 data. Pooled RMSE falls from 0.06605 for raw commands to 0.04563 for diagonal-affine and 0.03009 for coupled-affine calibration. The nested diagonal-linear and coupled-linear fits obtain 0.04828 and 0.03517, respectively. Coupled linear therefore outperforms diagonal affine without an intercept, isolating cross-axis response. Coupled-affine calibration reduces error by 54.45% versus raw commands and 34.07% versus diagonal-affine calibration. Its three held-out-session values are 0.02888, 0.03158, and 0.02974, and its 30-trial fit is within 3.33% of the 122-trial dense reference. The gain is concentrated laterally, where RMSE falls from 0.08459 to 0.02569 and the yaw-to-lateral coefficient remains 0.119–0.129 across holdouts. These data establish passive hardware model need rather than active hardware efficiency.

The synthetic benchmark isolates the acquisition rule (Fig. 5b). CalibAgent reaches the joint target in 18.67 trials, versus 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 without task weighting. The respective savings are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%; all paired tests remain significant after Holm correction. The primary LHS contrast saves 12.20 trials [10.73, 13.73]. Horizon-end active RMSE is 0.00846, and future-observation 95% predictive coverage is 94.14%.

Realized task error shows the same ordering (Fig. 5d). At budgets 18, 24, and 30, task-IVR RMSE is 0.02143, 0.01684, and 0.01420. Its paired improvement is 0.00392–0.00466 over D-optimal and 0.00545–0.00881 over no-task IVR; all six 95% intervals exclude zero.

Frozen designs remain effective under navigation-family reweighting (Fig. 5e): forward- and right-heavy tests retain positive intervals against every control, while the left-heavy D-optimal contrast becomes positive by 30 trials. A uniform safe-envelope measure reverses the ordering at 24 trials; task-IVR error is 3.21 times its declared-task error (Fig. 5f). Deployment support expansion therefore requires an updated task measure or a general-purpose design.

The uncertainty gate binds in all 60 active conditions. Task IVR requires 18, 18, and 20 trials across the three families, versus 19, 19, and 31 for D-optimal and 23, 23, and 31 without task weights (Fig. 5c). A disjoint 1,024-command audit gives crossing RMSE 0.01796–0.02315 and axiswise 95% coverage 0.917–0.955. Task IVR also saves at least 3.33 trials over D-optimal across nine predeclared gate pairs. Supplying exact heteroscedastic candidate noise changes its crossing from 20 to 21 trials while preserving the 11-trial advantage over both principal controls.

In closed-loop simulation (Fig. 3e), raw-to-calibrated RMSE is 0.15199 to 0.10635 for affine distortion, 0.15054 to 0.10282 for dead-zone distortion, 0.10741 to 0.09742 for low friction with payload and COM offset, and 0.09554 to 0.07849 on rough terrain with payload and COM offset. Relative reductions span 9.30–31.70%. The corresponding paired absolute-improvement intervals are [0.04290, 0.04780], [0.04495, 0.05002], [0.00981, 0.01017], and [0.01477, 0.01889]. All 20 paired seeds improve in each scenario.

B. Stopping, Safety, and Shift Recovery

The stopping test records no premature stop across 60 trajectories; median and p95 overshoot are two trials. The hard filter rejects 300/300 declared hazards, 0/20 safe controls, and all 160 sampled-state faults. In physics, the 50 Hz monitor's maximum abort interval is 20 ms. These software and simulator checks are not physical safety certification.

The stationary-monitor experiment produces 0/120 sequence alarms over 12,000 valid trials; the two-sided exact 95% upper bound is 0.0303. Under shift, detection is 143/144 for the combined friction, load, and gain change and the gain-and-coupling change, and 144/144 for the physical-and-nonlinear change and the load, COM, and gain change. The lowest exact 95% detection-rate bound is [0.9619, 0.9998]. CalibAgent recovers in 141/144 friction-plus-payload trials and 144/144 for the other shifts; its lowest exact interval is [0.9403, 0.9957]. Worst p95 detection and recovery delays are 3.9 and 9 trials, or 10.1 s of monitor-command time and 46.8 s for adaptation-plus-validation cycles under the fixed schedule.

CalibAgent improves early RMSE over passive updating by 0.00936 for the combined friction, load, and gain change, 0.01053 for the gain-and-coupling change, 0.01847 for the physical-and-nonlinear change, and 0.01192 for the load, COM, and gain change (Fig. 4f). Their paired 95% intervals are [0.00811, 0.01065], [0.00961, 0.01140], [0.01774, 0.01917], and [0.01092, 0.01291]. Its terminal RMSE ranges from 0.09762 to 0.11227; every upper interval endpoint is

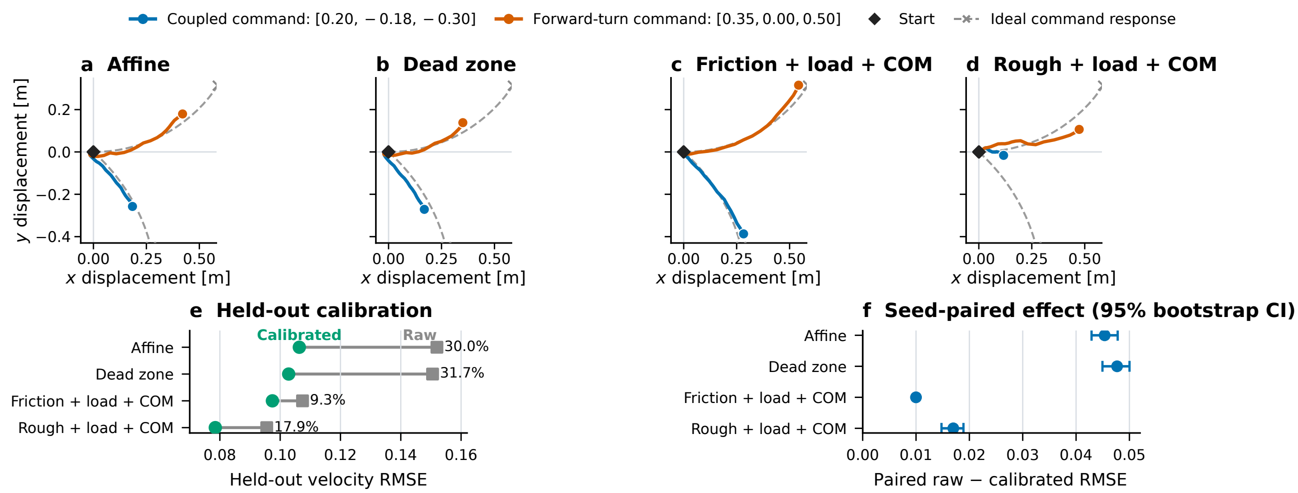


Fig. 3. Closed-loop simulation responses and calibration. (a)–(d) Body-displacement trajectories for coupled and forward-turn validation commands in a representative paired run; colored curves are measured, dashed gray curves and crosses are the ideal response to the declared command/timing, and diamonds mark the common start. (e) Raw and calibrated held-out velocity RMSE across 20 paired seeds; labels give relative reduction. (f) Seed-paired absolute improvement with 95% 4,000-sample bootstrap intervals. The trajectories are qualitative examples; all inference uses paired summaries specified before evaluation.

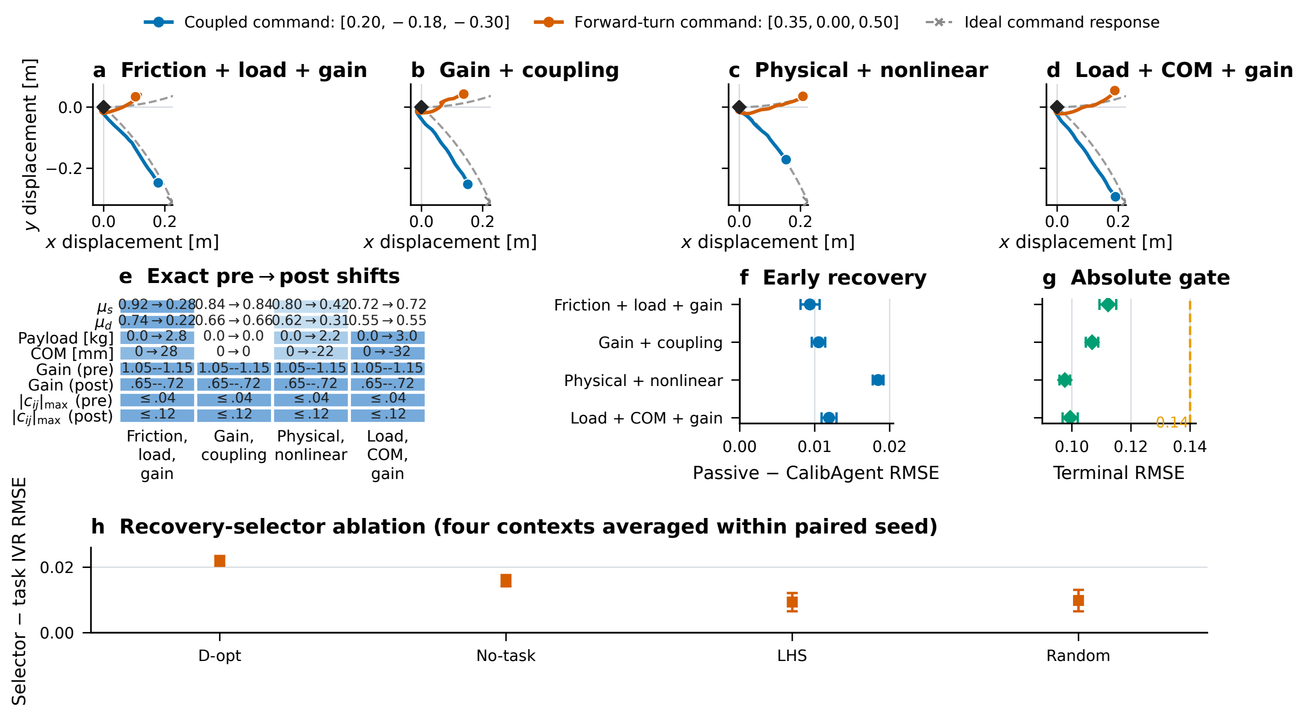


Fig. 4. Held-out simulator shifts and active recovery. (a)–(d) Measured body-displacement trajectories and dashed ideal command responses for the same validation commands in a representative paired run. (e) Exact pre→post static and dynamic friction, payload, fore–aft COM, per-axis gain range, and maximum off-axis coupling. Blue cells identify altered quantities, and every cell gives the exact pre- or post-shift value. The mixed condition additionally introduces dead zones, saturation, lag, and observation noise. (f) CalibAgent improvement over passive updating in the pre-specified early window. (g) CalibAgent terminal RMSE and the declared 0.14 gate. (h) Recovery-selector ablation, averaging the four contexts within each of 30 paired seeds; positive values favor task IVR. Error bars are paired 95% 4,000-sample bootstrap intervals. Recovery inference pools two disjoint 72-seed blocks per shift; stationary false alarms are evaluated separately over 120 sequences and 12,000 trials.

below 0.116 and the declared 0.14 absolute gate (Fig. 4g). These estimates pool both independent blocks and retain the fixed-model, passive-update, and CalibAgent pairing within every seed.

The selector ablation isolates acquisition (Fig. 4h). Task IVR lowers early RMSE by 0.02192 [0.02046, 0.02343] versus D-optimal, 0.01583 [0.01418, 0.01761] versus no-task IVR, 0.00937 [0.00655, 0.01212] versus LHS, and 0.00986 [0.00654, 0.01309] versus random selection. With no serious event, this establishes that early recovery depends on task-

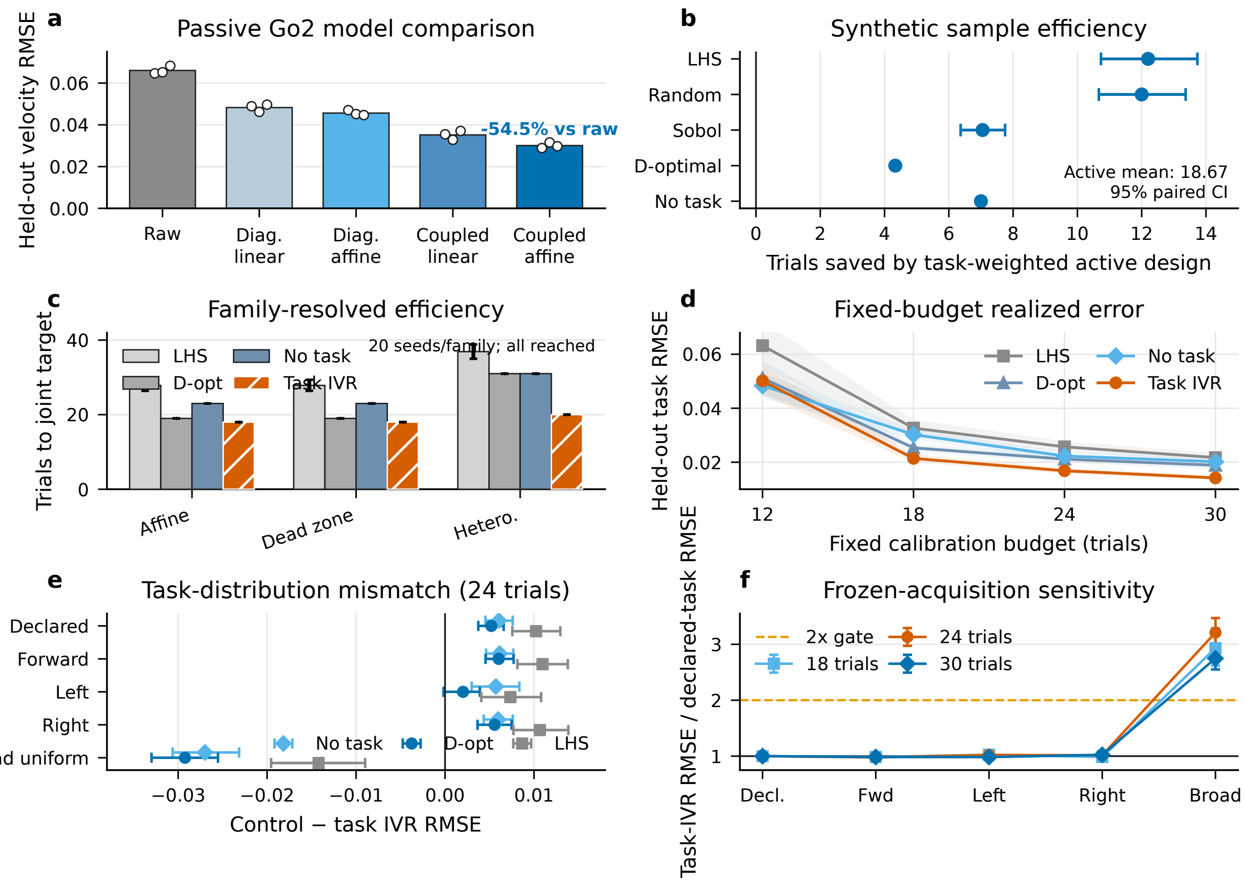


Fig. 5. Calibration evidence. (a) Passive real-Go2 held-out RMSE for raw, diagonal-linear, diagonal-affine, coupled-linear, and coupled-affine commands. (b) Paired trials saved by task IVR; bars are 95% seed-bootstrap intervals. (c) Family-resolved time to the same joint target; every method reaches it in all 20 seeds per family. (d) Realized held-out task RMSE at matched trial budgets; bands are 95% paired-seed bootstrap intervals. (e) Frozen-acquisition performance under task-distribution mismatch at 24 trials; positive control-minus-task-IVR values favor task IVR. (f) Task-IVR error relative to the declared measure; the uniform safe-envelope negative control crosses the pre-specified $2\times$ sensitivity gate.

directed commands rather than detection and updating alone.

C. Navigation Consequence

CalibAgent succeeds in 72/72 episodes on five maps and 70/72 on weighted arc, while raw commands succeed in 0/72 or 1/72 (Fig. 6a). Every CalibAgent collision count is zero. The failure-aware 60-s capped-time gain over raw has 95% lower endpoints of 23.45–32.68 s (Fig. 6b). At the same 12-trial budget, task IVR reduces command-validation RMSE over D-optimal by 9.98–12.28% on every map (Fig. 6c); its advantage over no-task IVR is positive on five maps and unresolved on S-bend.

The worst CalibAgent/dense capped-time ratio upper bound is 1.0736, below the 1.25 margin, and every paired success-difference lower bound exceeds -0.05 (Fig. 6d). The two weighted-arc timeouts are neither collisions nor serious events. Navigation excludes calibration: CalibAgent executes 31.2 s of acquisition commands versus 78.0 s for dense, or 52.0 versus 98.8 s after common validation commands are included. Across 432 episodes it records 10 non-serious safety events and no serious event. Identical seeds, policy, planner, interlock, maps, and navigation logic isolate calibration in this simulator pipeline.

VI. Discussion and Limitations

The experiments separate model need, task-directed acquisition, and downstream navigation. No-task and D-optimal controls isolate task weighting; noise and gate sensitivities preserve the ordering. Reweighting within the navigation family retains the benefit, whereas safe-envelope support expansion does not, so the task measure must change with deployment support. Dense and matched-budget navigation controls establish downstream noninferiority, while the raw arm exposes the cost of an uncalibrated interface.

The joint crossing is uncertainty-limited in all 60 active conditions, so the measured efficiency is faster acquisition of posterior confidence. Independent 400- and 1,024-command grids audit accuracy and predictive coverage; trace replay separately verifies stopping logic. Shift evaluation likewise separates four-command commissioning, 12,000 stationary monitor trials, shifted detection, and the fixed early-recovery window.

Hardware measurements are passive, same-day data from one Go2; active selection, recovery, and navigation use

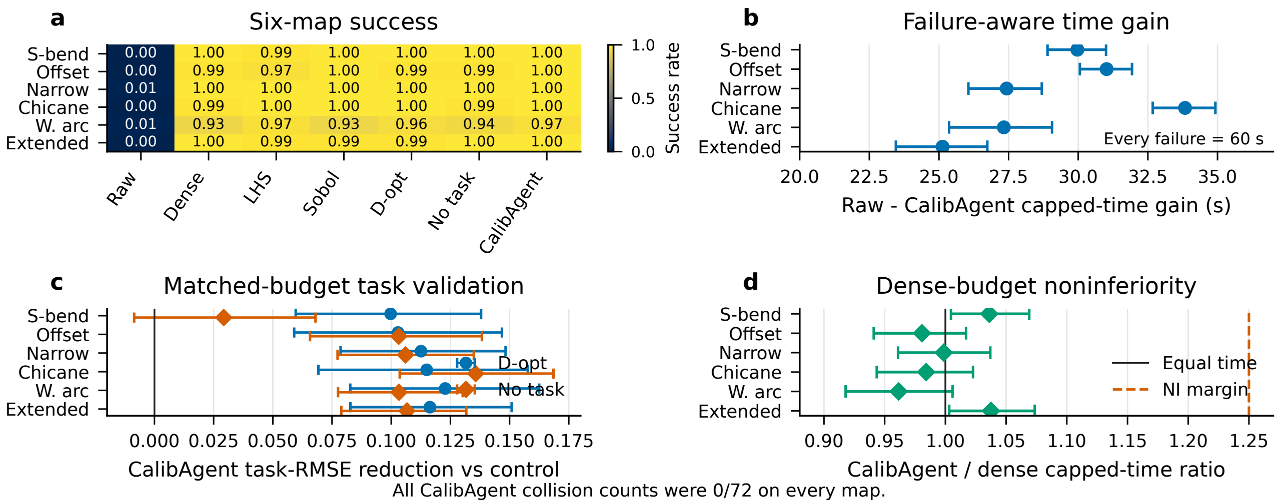


Fig. 6. Held-out simulator navigation (72 seeds/cell). (a) Success rate. (b) Paired raw-minus-CALIBAGENT gain in the 60-s capped-time estimand, which retains every failure as 60 s. (c) Matched-budget validation-RMSE reduction from task IVR relative to D-optimal and no-task IVR; bars are paired 95% intervals. (d) CALIBAGENT/dense capped-time ratios and the declared 1.25 noninferiority margin. Every CALIBAGENT collision count is 0/72.

a pinned simulator and fixed policy. The compact basis targets steady response, not arbitrary history dependence, gait switching, or rapid transients. Active hardware evaluation must apply the same mechanism under isolated and compound interface, load, center-of-mass, and friction changes, with weighted-arc and offset-slalom navigation. That evidence is required for physical closed-loop and sim-to-real claims.

VII. CONCLUSION

CALIBAGENT directs calibration trials toward task-relevant predictive uncertainty and composes coupled Bayesian regression with hard safety, validated stopping, bounded inversion, and latched recovery. Passive Go2 data establish model need; controlled synthetic and Isaac Lab experiments establish sample efficiency, early recovery, and downstream navigation. Active hardware execution is the remaining test of physical closed-loop transfer.

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